

IAMReX: an adaptive framework for the multiphase flow and fluid-particle interaction problems

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Summary

IAMReX is an adaptive C++ solver designed for multiphase flow and fluid-particle interaction problems. It is built in an object-oriented style and capable of high-performance massively parallel computing for complex systems (e.g., gas-fluid interaction, clusters of particles).

The original goal of IAMReX is to extend the capabilities of IAMR codes (Almgren et al., 1998), which only use a density-based solver to capture the diffuse interface of the two-phase flow. IAMReX offers a level set (LS) method and reinitialization techniques for accurately capturing the two-phase interface (Zeng et al., 2022), which increase the robustness of simulations with high Reynolds numbers (Zeng et al., 2023). In IAMR codes, particles are only treated as passive scalars. In IAMReX, however, both passive scalars and fully resolved particle cases are supported. To simulate the resolved particles, IAMReX employs the multidirect forcing immersed boundary method (Li et al., 2024). The associated Lagrangian markers used to resolve fluid-particle interfaces only exist on the finest-level grid, which greatly reduces memory cost. Both the subcycling and non-subcycling time advancement methods are implemented, and these methods help to decouple the time advancement at different levels. Like its predecessor, the IAMR library, IAMReX is also a publicly accessible platform designed specifically for developing massively parallel block-structured adaptive mesh refinement (BSAMR) applications. With the help of the AMReX framework, both IAMR and IAMReX support hybrid parallelization using either pure MPI or MPI & OpenMP for multicore machines (Zhang et al., 2019), and we have ensured that new features introduced in IAMReX do not compromise this cross-platform support.

The IAMReX code has undergone considerable development since 2023 and gained a few new contributors in the past two years. Although the projection-based flow solver is inherited from IAMR, IAMReX has added over 3,000 lines of new code, introduced 10 new test cases, and contributed approximately 60 new commits on GitHub. The key differences between IAMR and IAMReX are summarized in this [webpage](#). The versatility, accuracy, and efficiency of the present IAMReX framework are demonstrated by simulating two-phase flow and fluid-particle

45 interaction problems with various types of kinematic constraints. We carefully designed the
46 document such that users can easily compile and run cases. Input files, profiling scripts, and
47 raw post-processing data are also available for reproducing all results.

48 Scholarly effort

49 IAMReX's scholarly effort lies in its transformation from a standard incompressible flow solver
50 into a unified, high-performance, multi-physics framework for complex interfacial and particulate
51 flows. This is achieved through several key contributions. First, IAMReX introduces a modular
52 architecture that supports distinct and advanced physics modules within a single platform.
53 This includes a sharp-interface model for immiscible two-phase flows using an advanced level
54 set method, complete with conservative advection and mass-preserving reinitialization schemes.
55 Simultaneously, it offers a robust framework for particle-resolved fluid-structure interaction
56 (FSI) by implementing a Diffused Immersed Boundary Method (DIBM). Second, a significant
57 effort was dedicated to developing a comprehensive system for particle-resolved simulations.
58 This system features a DIBM implementation for handling the motion of 6-DOF rigid bodies
59 without requiring mesh conformity. It is further enhanced with a particle-particle collision
60 model using spatial hashing for efficient detection and a repulsive potential model for resolution.
61 Last but not least, IAMReX leverages the GPU acceleration capabilities to ensure efficiency for
62 particle-resolved simulations. All major particle operations, including fluid-solid coupling, force
63 spreading, and collision detection, are designed to be GPU-compatible. A key optimization
64 for memory efficiency involves constraining the Lagrangian markers used for the fluid-particle
65 interface to exist only on the finest grid level.

66 Relation to previous work

67 IAMReX is built upon the foundations established in three of our previous publications. The
68 implementation of the level set method is adapted from Zeng et al. (Zeng et al., 2022),
69 with a key distinction: IAMReX employs a semi-staggered grid rather than a collocated grid,
70 which strengthens the pressure-velocity coupling during the solution process. Furthermore, the
71 Diffused Immersed Boundary Method is derived from Li et al. (Li et al., 2024); while the original
72 work was limited to CPU-based parallelism and a basic particle collision model, IAMReX has
73 been tested on GPUs for select cases and is designed to accommodate more advanced collision
74 models. Finally, our approach to optimizing computational efficiency through the Adaptive
75 Mesh Refinement (AMR) technique, particularly the selection of simulation parameters, was
76 informed by the guidance provided in Liu et al. (Liu et al., 2025).

77 Statement of need

78 IAMReX is suitable for modeling multiphase flow problems and fluid-structure interaction
79 problems. Its level set-based interface capturing technique can be beneficial for researchers
80 studying phenomena such as wind over waves, breaking waves, and simulating the formation
81 and disappearance of bubbles and droplets. Additionally, the immersed boundary method along
82 with the collision models can parallelly resolve large-scale particles and capture their motions.
83 Researchers working on studies of biological particle aggregation, sandstorms, wind erosion of
84 ground surfaces, and seawater erosion of riverbeds are also among the target audience for this
85 software.

86 State of the field

87 We made great efforts to simulate more complex multiphase flows at higher resolution using
88 IAMReX. One effort is to combine the AMR technique with the multidirect forcing immersed

boundary method to resolve particles only on the finest-level grid. It significantly reduces the grid requirements for particle-resolved simulations compared with commonly used uniform grid solvers [Incompact3d](#), [CaNS](#), and [CP3d](#). Additionally, we utilized a subcycling technique to alleviate the time step constraint on coarser levels. It minimizes the total number of time steps needed for time advancement compared with the non-subcycling technique used in other AMR-related packages, such as [IBAMR](#), [basilisk](#), and [incflo](#).

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